Flammenkamp's Constant

Self-organization, attractors, and the limits of proof

What is the asymptotic density of live cells in a random GoL soup?

2.87115 %

empirical constant

Achim Flammenkamp

large-scale simulations, 1994–2004

Turing-complete system

GoL computes anything a UTM can

No analytic proof

undecidability is the hard barrier

Random Soups Converge

01

Initial soup

Each cell alive with probability $p_0 \in (0,1)$, independently. Maximum disorder, maximum entropy.

02

Transient dynamics

Local clusters form, interact, annihilate. Gliders, still lifes and oscillators emerge from noise.

03

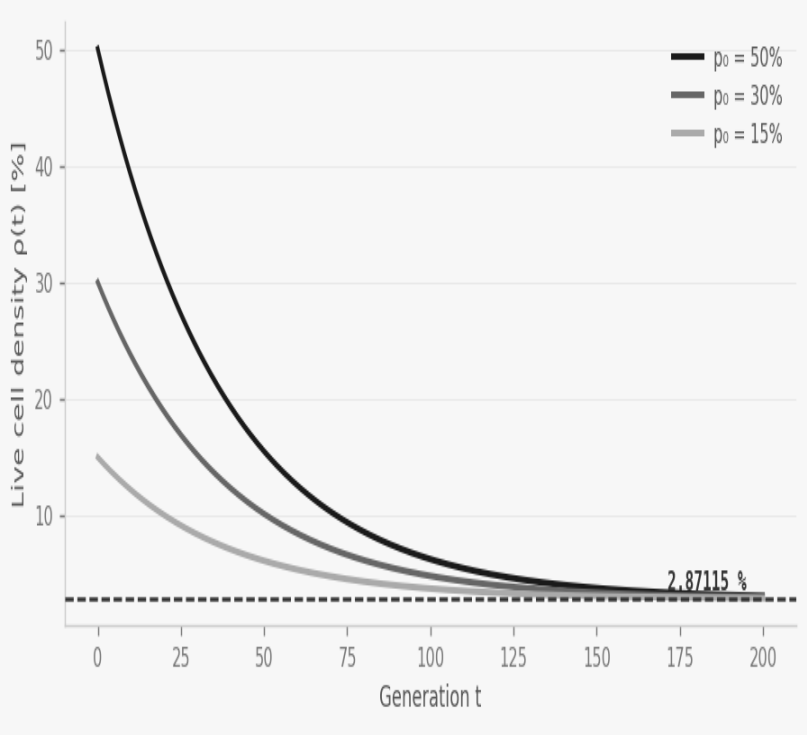
Quiescent state

The grid settles into isolated stable objects (ash). No further global evolution is possible.

04

Attractor density

Regardless of p_0 (over a wide range), the final live-cell density converges to the same value.



Density ρ(t) from three initial conditions → Flammenkamp attractor.
Approximate model: $\rho(t) \approx \rho_\infty + (\rho_0 - \rho_\infty)\, e^{-t/\tau}$

$\rho_\infty = 2.87115\ \%$

2.87115

%

≈ 1/e (e = 2.718...)

10^8 random soups

grids up to $10^4 \times 10^4$

Flammenkamp, 1994–2004

Universal attractor

ρ_∞ is independent of p_0 for a wide range ($\sim 0.10 < p_0 < 0.90$). This is the signature of a thermodynamic fixed point.

$$\forall\ p_0 \in (0.1,\ 0.9):\quad \rho(t\to\infty) \to \rho_\infty$$

Exponential relaxation

Density relaxes exponentially to ρ_∞ with time-scale $\tau_L \sim L^z$, $z = 1.66 \pm 0.05$ (Bagnoli et al., 1993).

$$\rho(t) \approx \rho_\infty + (\rho_0 - \rho_\infty)\, e^{-t/\tau_L}$$

Spatial correlations

Correlation length $\xi(t) \sim t^{1/z}$ before saturation. Cells are NOT independent — mean-field fails quantitatively.

$$\xi(t) \sim t^{1/z}, \qquad z \approx 1.66$$

Why Mean-Field Fails Here

Mean-field approximation

Assume cells are independent: P(alive) = ρ(t).

Birth rule: dead cell with exactly 3 live neighbors.

$$P(birth) = C(8,3)\cdot\rho^3(1-\rho)^5 = 56\rho^3(1-\rho)^5$$

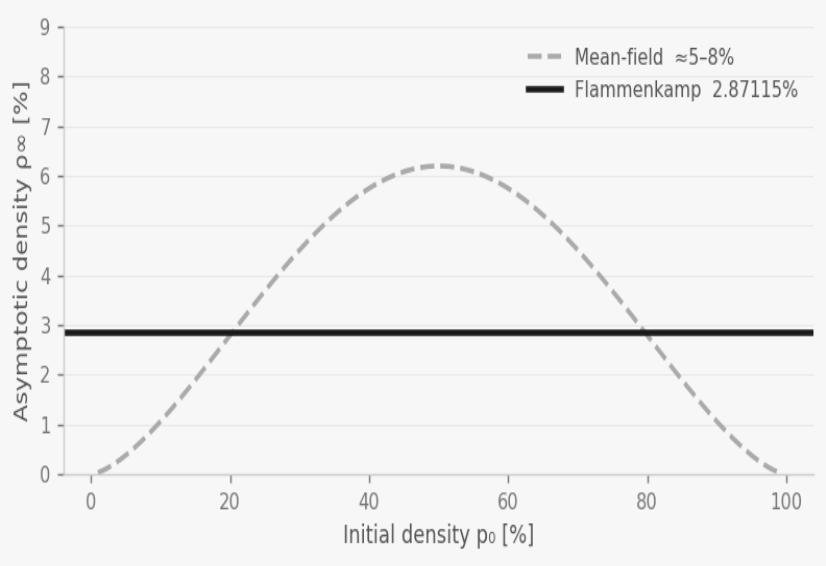
Survival: live cell with 2 or 3 live neighbors.

$$P(survive) = C(8,2)\rho^2(1-\rho)^6 + 56\rho^3(1-\rho)^5$$

Fixed point: $\rho^* = P(birth) + \rho\cdot P(survive) - \rho$

MF prediction: $\rho^* \approx 5$–8% (depends on p_0)

MF predicts ~5–8% vs. **Flammenkamp 2.87%**



Asymptotic density vs. p_0. MF overestimates ρ_∞ because it ignores spatial correlations and cluster structure.

What mean-field ignores

→ Spatial clusters: still lifes and oscillators are highly correlated structures.

→ Correlations extend over ξ(t) ~ t^{0.6}: cells are far from independent.

→ Ash objects are large, sparse and stable — not capturable by local ρ alone.

Why No Analytic Proof Exists

1

GoL is Turing-complete

Any computable function can be encoded as a GoL initial configuration. Gliders act as wires; collisions implement logic gates (AND, OR, NOT).

2

Turing-completeness $\Longrightarrow$ Halting Problem applies

For any Turing-complete system, there is no general algorithm to determine whether a given computation terminates. Rice's theorem generalizes this.

3

Long-term behavior is undecidable

Given pattern P, it is formally undecidable whether P will ever stabilize, or which attractor it will reach. Proved as a corollary of the Halting Problem.

4

ρ_∞ cannot be derived analytically — the barrier is fundamental

Computing ρ_∞ would require predicting the fate of every configuration — equivalent to solving the Halting Problem. This is not a gap in technique; it is a formal impossibility.

Flammenkamp's Constant as a Physical Object

Fixed-point attractor

Statistical mechanics

ρ_∞ is a fixed point of the density map $T: \rho \mapsto T(\rho)$. The system flows toward it from any initial condition, like a thermodynamic equilibrium state.

$T(\rho_\infty) = \rho_\infty$

Order parameter

Phase transitions

$\rho(t)$ plays the role of an order parameter. GoL undergoes an order-disorder transition from soup to ash. ρ_∞ is the equilibrium value of the absorbing state.

$\Phi = \rho_\infty$ (absorbing state)

Renormalization group

RG flow

Mean-field is the zeroth-order RG approximation. ρ_∞ is the true fixed point of the full RG flow; MF truncation gives the wrong answer because correlations are relevant.

$\rho^*_{MF} \approx 5\text{–}8\% \neq \rho_\infty = 2.87\%$

Self-organized criticality

BTW sandpile

Like the Bak-Tang-Wiesenfeld sandpile, GoL self-organizes to a critical state without tuning a control parameter. ρ_∞ is the critical density of the absorbing-state transition.

no tuning parameter → ρ_∞

Finite-size scaling

Spin systems near Tc

$\tau_L \sim L^{1.66}$ is the relaxation timescale. This power-law FSS is structurally identical to spin models near their critical temperature.

$\tau_L \propto L^z, \quad z = 1.66 \pm 0.05$

Undecidability as phase

Edge of chaos (Wolfram Class IV)

Class IV CAs live at the boundary of order and chaos. GoL is Class IV: its long-term behavior is undecidable. ρ_∞ exists and is measurable, but is not derivable — like Chaitin's Ω.

Class IV ↔ undecidable long-run

Key takeaway: $\rho_\infty = 2.87115\%$ is an empirical constant like α or G — its existence is clear, its value is measured, but its derivation lies beyond current mathematics.